

NCERT CLASS 12

60 Questions Per Topic

Based on the refined 2026–27 curriculum map created for this project.

Each curriculum topic below contains exactly 60 questions: understanding, examples/classification, comparison, application, reasoning/error analysis, and higher-order activity.

These are question-bank items derived from the topic descriptions in the supplied curriculum map. Validate against the exact school/textbook edition before high-stakes use.

MATHEMATICS — 1. Relations and Functions

Coverage: ■ Relations; one-one/onto; composition; inverse functions.

Understanding

1. What is ■ Relations, and what are its main features? Use a simple example.
2. What is one-one/onto, and what are its main features? Use a school-life example.
3. What is composition, and what are its main features? Use a real-world example.
4. What is inverse functions., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Relations, and what are its main features? Explain in your own words.
6. What is one-one/onto, and what are its main features? Give a step-by-step response.
7. What is composition, and what are its main features? Mention one common misconception.
8. What is inverse functions., and what are its main features? State one practical application.
9. What is ■ Relations, and what are its main features? Justify your response.
10. What is one-one/onto, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to composition and explain each. Use a simple example.
12. Give two examples related to inverse functions. and explain each. Use a school-life example.
13. Give two examples related to ■ Relations and explain each. Use a real-world example.
14. Give two examples related to one-one/onto and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to composition and explain each. Explain in your own words.
16. Give two examples related to inverse functions. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Relations and explain each. Mention one common misconception.
18. Give two examples related to one-one/onto and explain each. State one practical application.
19. Give two examples related to composition and explain each. Justify your response.
20. Give two examples related to inverse functions. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Relations related to one-one/onto? Compare the two ideas. Use a simple example.
22. How is one-one/onto related to composition? Compare the two ideas. Use a school-life example.
23. How is composition related to inverse functions.? Compare the two ideas. Use a real-world example.
24. How is inverse functions. related to ■ Relations? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Relations related to one-one/onto? Compare the two ideas. Explain in your own words.
26. How is one-one/onto related to composition? Compare the two ideas. Give a step-by-step response.
27. How is composition related to inverse functions.? Compare the two ideas. Mention one common misconception.
28. How is inverse functions. related to ■ Relations? Compare the two ideas. State one practical application.
29. How is ■ Relations related to one-one/onto? Compare the two ideas. Justify your response.
30. How is one-one/onto related to composition? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply composition to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply inverse functions. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Relations to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply one-one/onto to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply composition to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply inverse functions. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Relations to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply one-one/onto to a realistic situation. What would you do or calculate? State one practical application.
39. Apply composition to a realistic situation. What would you do or calculate? Justify your response.
40. Apply inverse functions. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Relations. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of one-one/onto. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of composition. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of inverse functions.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Relations. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of one-one/onto. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of composition. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of inverse functions.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Relations. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of one-one/onto. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show composition. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show inverse functions.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Relations. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show one-one/onto. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show composition. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show inverse functions.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Relations. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show one-one/onto. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show composition. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show inverse functions.. Include the relevant rule or principle.

MATHEMATICS — 2. Inverse Trigonometric Functions

Coverage: ■ Properties; principal values; graphs; identities.

Understanding

1. What is ■ Properties, and what are its main features? Use a simple example.
2. What is principal values, and what are its main features? Use a school-life example.
3. What is graphs, and what are its main features? Use a real-world example.
4. What is identities., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Properties, and what are its main features? Explain in your own words.
6. What is principal values, and what are its main features? Give a step-by-step response.
7. What is graphs, and what are its main features? Mention one common misconception.
8. What is identities., and what are its main features? State one practical application.
9. What is ■ Properties, and what are its main features? Justify your response.
10. What is principal values, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to graphs and explain each. Use a simple example.
12. Give two examples related to identities. and explain each. Use a school-life example.
13. Give two examples related to ■ Properties and explain each. Use a real-world example.
14. Give two examples related to principal values and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to graphs and explain each. Explain in your own words.
16. Give two examples related to identities. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Properties and explain each. Mention one common misconception.
18. Give two examples related to principal values and explain each. State one practical application.
19. Give two examples related to graphs and explain each. Justify your response.
20. Give two examples related to identities. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Properties related to principal values? Compare the two ideas. Use a simple example.
22. How is principal values related to graphs? Compare the two ideas. Use a school-life example.
23. How is graphs related to identities.? Compare the two ideas. Use a real-world example.
24. How is identities. related to ■ Properties? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Properties related to principal values? Compare the two ideas. Explain in your own words.
26. How is principal values related to graphs? Compare the two ideas. Give a step-by-step response.
27. How is graphs related to identities.? Compare the two ideas. Mention one common misconception.
28. How is identities. related to ■ Properties? Compare the two ideas. State one practical application.
29. How is ■ Properties related to principal values? Compare the two ideas. Justify your response.
30. How is principal values related to graphs? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply graphs to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply identities. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Properties to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply principal values to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply graphs to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply identities. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Properties to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply principal values to a realistic situation. What would you do or calculate? State one practical application.
39. Apply graphs to a realistic situation. What would you do or calculate? Justify your response.
40. Apply identities. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Properties. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of principal values. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of graphs. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of identities.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Properties. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of principal values. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of graphs. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of identities.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Properties. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of principal values. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show graphs. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show identities.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Properties. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show principal values. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show graphs. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show identities.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Properties. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show principal values. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show graphs. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show identities.. Include the relevant rule or principle.

MATHEMATICS — 3. Matrices

Coverage: ■ Types; operations; transpose; symmetric/skew-symmetric; inverse.

Understanding

1. What is ■ Types, and what are its main features? Use a simple example.
2. What is operations, and what are its main features? Use a school-life example.
3. What is transpose, and what are its main features? Use a real-world example.
4. What is symmetric/skew-symmetric, and what are its main features? Show the idea with a diagram where appropriate.
5. What is inverse., and what are its main features? Explain in your own words.
6. What is ■ Types, and what are its main features? Give a step-by-step response.
7. What is operations, and what are its main features? Mention one common misconception.
8. What is transpose, and what are its main features? State one practical application.
9. What is symmetric/skew-symmetric, and what are its main features? Justify your response.
10. What is inverse., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Types and explain each. Use a simple example.
12. Give two examples related to operations and explain each. Use a school-life example.
13. Give two examples related to transpose and explain each. Use a real-world example.
14. Give two examples related to symmetric/skew-symmetric and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to inverse. and explain each. Explain in your own words.
16. Give two examples related to ■ Types and explain each. Give a step-by-step response.
17. Give two examples related to operations and explain each. Mention one common misconception.
18. Give two examples related to transpose and explain each. State one practical application.
19. Give two examples related to symmetric/skew-symmetric and explain each. Justify your response.
20. Give two examples related to inverse. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Types related to operations? Compare the two ideas. Use a simple example.
22. How is operations related to transpose? Compare the two ideas. Use a school-life example.
23. How is transpose related to symmetric/skew-symmetric? Compare the two ideas. Use a real-world example.
24. How is symmetric/skew-symmetric related to inverse.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is inverse. related to ■ Types? Compare the two ideas. Explain in your own words.
26. How is ■ Types related to operations? Compare the two ideas. Give a step-by-step response.
27. How is operations related to transpose? Compare the two ideas. Mention one common misconception.
28. How is transpose related to symmetric/skew-symmetric? Compare the two ideas. State one practical application.
29. How is symmetric/skew-symmetric related to inverse.? Compare the two ideas. Justify your response.
30. How is inverse. related to ■ Types? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Types to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply operations to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply transpose to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply symmetric/skew-symmetric to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply inverse. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Types to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply operations to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply transpose to a realistic situation. What would you do or calculate? State one practical application.
39. Apply symmetric/skew-symmetric to a realistic situation. What would you do or calculate? Justify your response.
40. Apply inverse. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Types. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of operations. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of transpose. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of symmetric/skew-symmetric. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of inverse.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Types. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of operations. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of transpose. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of symmetric/skew-symmetric. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of inverse.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Types. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show operations. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show transpose. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show symmetric/skew-symmetric. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show inverse.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Types. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show operations. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show transpose. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show symmetric/skew-symmetric. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show inverse.. Include the relevant rule or principle.

MATHEMATICS — 4. Determinants

Coverage: ■ Properties; minors/cofactors; adjoint/inverse; linear equations.

Understanding

1. What is ■ Properties, and what are its main features? Use a simple example.
2. What is minors/cofactors, and what are its main features? Use a school-life example.
3. What is adjoint/inverse, and what are its main features? Use a real-world example.
4. What is linear equations., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Properties, and what are its main features? Explain in your own words.
6. What is minors/cofactors, and what are its main features? Give a step-by-step response.
7. What is adjoint/inverse, and what are its main features? Mention one common misconception.
8. What is linear equations., and what are its main features? State one practical application.
9. What is ■ Properties, and what are its main features? Justify your response.
10. What is minors/cofactors, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to adjoint/inverse and explain each. Use a simple example.
12. Give two examples related to linear equations. and explain each. Use a school-life example.
13. Give two examples related to ■ Properties and explain each. Use a real-world example.
14. Give two examples related to minors/cofactors and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to adjoint/inverse and explain each. Explain in your own words.
16. Give two examples related to linear equations. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Properties and explain each. Mention one common misconception.
18. Give two examples related to minors/cofactors and explain each. State one practical application.
19. Give two examples related to adjoint/inverse and explain each. Justify your response.
20. Give two examples related to linear equations. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Properties related to minors/cofactors? Compare the two ideas. Use a simple example.
22. How is minors/cofactors related to adjoint/inverse? Compare the two ideas. Use a school-life example.
23. How is adjoint/inverse related to linear equations.? Compare the two ideas. Use a real-world example.
24. How is linear equations. related to ■ Properties? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Properties related to minors/cofactors? Compare the two ideas. Explain in your own words.
26. How is minors/cofactors related to adjoint/inverse? Compare the two ideas. Give a step-by-step response.
27. How is adjoint/inverse related to linear equations.? Compare the two ideas. Mention one common misconception.
28. How is linear equations. related to ■ Properties? Compare the two ideas. State one practical application.
29. How is ■ Properties related to minors/cofactors? Compare the two ideas. Justify your response.
30. How is minors/cofactors related to adjoint/inverse? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply adjoint/inverse to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply linear equations. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Properties to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply minors/cofactors to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply adjoint/inverse to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply linear equations. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Properties to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply minors/cofactors to a realistic situation. What would you do or calculate? State one practical application.
39. Apply adjoint/inverse to a realistic situation. What would you do or calculate? Justify your response.
40. Apply linear equations. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Properties. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of minors/cofactors. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of adjoint/inverse. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of linear equations.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Properties. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of minors/cofactors. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of adjoint/inverse. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of linear equations.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Properties. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of minors/cofactors. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show adjoint/inverse. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show linear equations.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Properties. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show minors/cofactors. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show adjoint/inverse. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show linear equations.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Properties. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show minors/cofactors. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show adjoint/inverse. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show linear equations.. Include the relevant rule or principle.

MATHEMATICS — 5. Continuity and Differentiability

Coverage: ■ Continuity; derivatives; chain/log differentiation; mean value theorems.

Understanding

1. What is ■ Continuity, and what are its main features? Use a simple example.
2. What is derivatives, and what are its main features? Use a school-life example.
3. What is chain/log differentiation, and what are its main features? Use a real-world example.
4. What is mean value theorems., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Continuity, and what are its main features? Explain in your own words.
6. What is derivatives, and what are its main features? Give a step-by-step response.
7. What is chain/log differentiation, and what are its main features? Mention one common misconception.
8. What is mean value theorems., and what are its main features? State one practical application.
9. What is ■ Continuity, and what are its main features? Justify your response.
10. What is derivatives, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to chain/log differentiation and explain each. Use a simple example.
12. Give two examples related to mean value theorems. and explain each. Use a school-life example.
13. Give two examples related to ■ Continuity and explain each. Use a real-world example.
14. Give two examples related to derivatives and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to chain/log differentiation and explain each. Explain in your own words.
16. Give two examples related to mean value theorems. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Continuity and explain each. Mention one common misconception.
18. Give two examples related to derivatives and explain each. State one practical application.
19. Give two examples related to chain/log differentiation and explain each. Justify your response.
20. Give two examples related to mean value theorems. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Continuity related to derivatives? Compare the two ideas. Use a simple example.
22. How is derivatives related to chain/log differentiation? Compare the two ideas. Use a school-life example.
23. How is chain/log differentiation related to mean value theorems.? Compare the two ideas. Use a real-world example.
24. How is mean value theorems. related to ■ Continuity? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Continuity related to derivatives? Compare the two ideas. Explain in your own words.
26. How is derivatives related to chain/log differentiation? Compare the two ideas. Give a step-by-step response.
27. How is chain/log differentiation related to mean value theorems.? Compare the two ideas. Mention one common misconception.
28. How is mean value theorems. related to ■ Continuity? Compare the two ideas. State one practical application.
29. How is ■ Continuity related to derivatives? Compare the two ideas. Justify your response.
30. How is derivatives related to chain/log differentiation? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply chain/log differentiation to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply mean value theorems. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Continuity to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply derivatives to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply chain/log differentiation to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply mean value theorems. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Continuity to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply derivatives to a realistic situation. What would you do or calculate? State one practical application.
39. Apply chain/log differentiation to a realistic situation. What would you do or calculate? Justify your response.
40. Apply mean value theorems. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Continuity. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of derivatives. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of chain/log differentiation. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of mean value theorems.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Continuity. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of derivatives. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of chain/log differentiation. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of mean value theorems.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Continuity. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of derivatives. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show chain/log differentiation. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show mean value theorems.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Continuity. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show derivatives. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show chain/log differentiation. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show mean value theorems.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Continuity. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show derivatives. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show chain/log differentiation. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show mean value theorems.. Include the relevant rule or principle.

MATHEMATICS — 6. Application of Derivatives

Coverage: ■ Monotonicity; rate of change; maxima/minima; tangents/normals.

Understanding

1. What is ■ Monotonicity, and what are its main features? Use a simple example.
2. What is rate of change, and what are its main features? Use a school-life example.
3. What is maxima/minima, and what are its main features? Use a real-world example.
4. What is tangents/normals., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Monotonicity, and what are its main features? Explain in your own words.
6. What is rate of change, and what are its main features? Give a step-by-step response.
7. What is maxima/minima, and what are its main features? Mention one common misconception.
8. What is tangents/normals., and what are its main features? State one practical application.
9. What is ■ Monotonicity, and what are its main features? Justify your response.
10. What is rate of change, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to maxima/minima and explain each. Use a simple example.
12. Give two examples related to tangents/normals. and explain each. Use a school-life example.
13. Give two examples related to ■ Monotonicity and explain each. Use a real-world example.
14. Give two examples related to rate of change and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to maxima/minima and explain each. Explain in your own words.
16. Give two examples related to tangents/normals. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Monotonicity and explain each. Mention one common misconception.
18. Give two examples related to rate of change and explain each. State one practical application.
19. Give two examples related to maxima/minima and explain each. Justify your response.
20. Give two examples related to tangents/normals. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Monotonicity related to rate of change? Compare the two ideas. Use a simple example.
22. How is rate of change related to maxima/minima? Compare the two ideas. Use a school-life example.
23. How is maxima/minima related to tangents/normals.? Compare the two ideas. Use a real-world example.
24. How is tangents/normals. related to ■ Monotonicity? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Monotonicity related to rate of change? Compare the two ideas. Explain in your own words.
26. How is rate of change related to maxima/minima? Compare the two ideas. Give a step-by-step response.
27. How is maxima/minima related to tangents/normals.? Compare the two ideas. Mention one common misconception.
28. How is tangents/normals. related to ■ Monotonicity? Compare the two ideas. State one practical application.
29. How is ■ Monotonicity related to rate of change? Compare the two ideas. Justify your response.
30. How is rate of change related to maxima/minima? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply maxima/minima to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply tangents/normals. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Monotonicity to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply rate of change to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply maxima/minima to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply tangents/normals. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Monotonicity to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply rate of change to a realistic situation. What would you do or calculate? State one practical application.
39. Apply maxima/minima to a realistic situation. What would you do or calculate? Justify your response.
40. Apply tangents/normals. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Monotonicity. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of rate of change. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of maxima/minima. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of tangents/normals.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Monotonicity. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of rate of change. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of maxima/minima. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of tangents/normals.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Monotonicity. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of rate of change. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show maxima/minima. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show tangents/normals.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Monotonicity. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show rate of change. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show maxima/minima. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show tangents/normals.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Monotonicity. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show rate of change. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show maxima/minima. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show tangents/normals.. Include the relevant rule or principle.

MATHEMATICS — 7. Integrals

Coverage: ■ Indefinite/definite integration; substitution; standard methods; properties.

Understanding

1. What is ■ Indefinite/definite integration, and what are its main features? Use a simple example.
2. What is substitution, and what are its main features? Use a school-life example.
3. What is standard methods, and what are its main features? Use a real-world example.
4. What is properties., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Indefinite/definite integration, and what are its main features? Explain in your own words.
6. What is substitution, and what are its main features? Give a step-by-step response.
7. What is standard methods, and what are its main features? Mention one common misconception.
8. What is properties., and what are its main features? State one practical application.
9. What is ■ Indefinite/definite integration, and what are its main features? Justify your response.
10. What is substitution, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to standard methods and explain each. Use a simple example.
12. Give two examples related to properties. and explain each. Use a school-life example.
13. Give two examples related to ■ Indefinite/definite integration and explain each. Use a real-world example.
14. Give two examples related to substitution and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to standard methods and explain each. Explain in your own words.
16. Give two examples related to properties. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Indefinite/definite integration and explain each. Mention one common misconception.
18. Give two examples related to substitution and explain each. State one practical application.
19. Give two examples related to standard methods and explain each. Justify your response.
20. Give two examples related to properties. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Indefinite/definite integration related to substitution? Compare the two ideas. Use a simple example.
22. How is substitution related to standard methods? Compare the two ideas. Use a school-life example.
23. How is standard methods related to properties.? Compare the two ideas. Use a real-world example.
24. How is properties. related to ■ Indefinite/definite integration? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Indefinite/definite integration related to substitution? Compare the two ideas. Explain in your own words.
26. How is substitution related to standard methods? Compare the two ideas. Give a step-by-step response.
27. How is standard methods related to properties.? Compare the two ideas. Mention one common misconception.
28. How is properties. related to ■ Indefinite/definite integration? Compare the two ideas. State one practical application.
29. How is ■ Indefinite/definite integration related to substitution? Compare the two ideas. Justify your response.
30. How is substitution related to standard methods? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply standard methods to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply properties. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Indefinite/definite integration to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply substitution to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply standard methods to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply properties. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Indefinite/definite integration to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply substitution to a realistic situation. What would you do or calculate? State one practical application.
39. Apply standard methods to a realistic situation. What would you do or calculate? Justify your response.
40. Apply properties. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Indefinite/definite integration. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of substitution. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of standard methods. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of properties.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Indefinite/definite integration. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of substitution. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of standard methods. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of properties.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Indefinite/definite integration. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of substitution. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show standard methods. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show properties.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Indefinite/definite integration. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show substitution. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show standard methods. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show properties.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Indefinite/definite integration. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show substitution. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show standard methods. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show properties.. Include the relevant rule or principle.

MATHEMATICS — 8. Application of Integrals

Coverage: ■ Area under/between curves.

Understanding

1. What is ■ Area under/between curves., and what are its main features? Use a simple example.
2. What is ■ Area under/between curves., and what are its main features? Use a school-life example.
3. What is ■ Area under/between curves., and what are its main features? Use a real-world example.
4. What is ■ Area under/between curves., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Area under/between curves., and what are its main features? Explain in your own words.
6. What is ■ Area under/between curves., and what are its main features? Give a step-by-step response.
7. What is ■ Area under/between curves., and what are its main features? Mention one common misconception.
8. What is ■ Area under/between curves., and what are its main features? State one practical application.
9. What is ■ Area under/between curves., and what are its main features? Justify your response.
10. What is ■ Area under/between curves., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Area under/between curves. and explain each. Use a simple example.
12. Give two examples related to ■ Area under/between curves. and explain each. Use a school-life example.
13. Give two examples related to ■ Area under/between curves. and explain each. Use a real-world example.
14. Give two examples related to ■ Area under/between curves. and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to ■ Area under/between curves. and explain each. Explain in your own words.
16. Give two examples related to ■ Area under/between curves. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Area under/between curves. and explain each. Mention one common misconception.
18. Give two examples related to ■ Area under/between curves. and explain each. State one practical application.
19. Give two examples related to ■ Area under/between curves. and explain each. Justify your response.
20. Give two examples related to ■ Area under/between curves. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Area under/between curves. related to ■ Area under/between curves.? Compare the two ideas. Use a simple example.
22. How is ■ Area under/between curves. related to ■ Area under/between curves.? Compare the two ideas. Use a school-life example.
23. How is ■ Area under/between curves. related to ■ Area under/between curves.? Compare the two ideas. Use a real-world example.
24. How is ■ Area under/between curves. related to ■ Area under/between curves.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Area under/between curves. related to ■ Area under/between curves.? Compare the two ideas. Explain in your own words.
26. How is ■ Area under/between curves. related to ■ Area under/between curves.? Compare the two ideas. Give a step-by-step response.
27. How is ■ Area under/between curves. related to ■ Area under/between curves.? Compare the two ideas. Mention one common misconception.
28. How is ■ Area under/between curves. related to ■ Area under/between curves.? Compare the two ideas. State one practical application.
29. How is ■ Area under/between curves. related to ■ Area under/between curves.? Compare the two ideas. Justify your response.
30. How is ■ Area under/between curves. related to ■ Area under/between curves.? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Area under/between curves. to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply ■ Area under/between curves. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Area under/between curves. to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply ■ Area under/between curves. to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply ■ Area under/between curves. to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Area under/between curves. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Area under/between curves. to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply ■ Area under/between curves. to a realistic situation. What would you do or calculate? State one practical application.
39. Apply ■ Area under/between curves. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Area under/between curves. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Area under/between curves.. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of ■ Area under/between curves.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Area under/between curves.. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of ■ Area under/between curves.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Area under/between curves.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Area under/between curves.. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of ■ Area under/between curves.. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of ■ Area under/between curves.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Area under/between curves.. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of ■ Area under/between curves.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Area under/between curves.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Area under/between curves.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Area under/between curves.. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Area under/between curves.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Area under/between curves.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Area under/between curves.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Area under/between curves.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Area under/between curves.. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Area under/between curves.. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Area under/between curves.. Include the relevant rule or principle.

MATHEMATICS — 9. Differential Equations

Coverage: ■ Order/degree; general/particular solution; first-order equations.

Understanding

1. What is ■ Order/degree, and what are its main features? Use a simple example.
2. What is general/particular solution, and what are its main features? Use a school-life example.
3. What is first-order equations., and what are its main features? Use a real-world example.
4. What is ■ Order/degree, and what are its main features? Show the idea with a diagram where appropriate.
5. What is general/particular solution, and what are its main features? Explain in your own words.
6. What is first-order equations., and what are its main features? Give a step-by-step response.
7. What is ■ Order/degree, and what are its main features? Mention one common misconception.
8. What is general/particular solution, and what are its main features? State one practical application.
9. What is first-order equations., and what are its main features? Justify your response.
10. What is ■ Order/degree, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to general/particular solution and explain each. Use a simple example.
12. Give two examples related to first-order equations. and explain each. Use a school-life example.
13. Give two examples related to ■ Order/degree and explain each. Use a real-world example.
14. Give two examples related to general/particular solution and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to first-order equations. and explain each. Explain in your own words.
16. Give two examples related to ■ Order/degree and explain each. Give a step-by-step response.
17. Give two examples related to general/particular solution and explain each. Mention one common misconception.
18. Give two examples related to first-order equations. and explain each. State one practical application.
19. Give two examples related to ■ Order/degree and explain each. Justify your response.
20. Give two examples related to general/particular solution and explain each. Include the relevant rule or principle.

Comparison

21. How is first-order equations. related to ■ Order/degree? Compare the two ideas. Use a simple example.
22. How is ■ Order/degree related to general/particular solution? Compare the two ideas. Use a school-life example.
23. How is general/particular solution related to first-order equations.? Compare the two ideas. Use a real-world example.
24. How is first-order equations. related to ■ Order/degree? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Order/degree related to general/particular solution? Compare the two ideas. Explain in your own words.
26. How is general/particular solution related to first-order equations.? Compare the two ideas. Give a step-by-step response.
27. How is first-order equations. related to ■ Order/degree? Compare the two ideas. Mention one common misconception.
28. How is ■ Order/degree related to general/particular solution? Compare the two ideas. State one practical application.
29. How is general/particular solution related to first-order equations.? Compare the two ideas. Justify your response.
30. How is first-order equations. related to ■ Order/degree? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Order/degree to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply general/particular solution to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply first-order equations. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Order/degree to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply general/particular solution to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply first-order equations. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Order/degree to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply general/particular solution to a realistic situation. What would you do or calculate? State one practical application.
39. Apply first-order equations. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Order/degree to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of general/particular solution. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of first-order equations.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Order/degree. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of general/particular solution. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of first-order equations.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Order/degree. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of general/particular solution. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of first-order equations.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Order/degree. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of general/particular solution. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show first-order equations.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Order/degree. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show general/particular solution. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show first-order equations.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Order/degree. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show general/particular solution. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show first-order equations.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Order/degree. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show general/particular solution. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show first-order equations.. Include the relevant rule or principle.

MATHEMATICS — 10. Vector Algebra

Coverage: ■ Vectors; dot/cross product; geometric applications.

Understanding

1. What is ■ Vectors, and what are its main features? Use a simple example.
2. What is dot/cross product, and what are its main features? Use a school-life example.
3. What is geometric applications., and what are its main features? Use a real-world example.
4. What is ■ Vectors, and what are its main features? Show the idea with a diagram where appropriate.
5. What is dot/cross product, and what are its main features? Explain in your own words.
6. What is geometric applications., and what are its main features? Give a step-by-step response.
7. What is ■ Vectors, and what are its main features? Mention one common misconception.
8. What is dot/cross product, and what are its main features? State one practical application.
9. What is geometric applications., and what are its main features? Justify your response.
10. What is ■ Vectors, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to dot/cross product and explain each. Use a simple example.
12. Give two examples related to geometric applications. and explain each. Use a school-life example.
13. Give two examples related to ■ Vectors and explain each. Use a real-world example.
14. Give two examples related to dot/cross product and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to geometric applications. and explain each. Explain in your own words.
16. Give two examples related to ■ Vectors and explain each. Give a step-by-step response.
17. Give two examples related to dot/cross product and explain each. Mention one common misconception.
18. Give two examples related to geometric applications. and explain each. State one practical application.
19. Give two examples related to ■ Vectors and explain each. Justify your response.
20. Give two examples related to dot/cross product and explain each. Include the relevant rule or principle.

Comparison

21. How is geometric applications. related to ■ Vectors? Compare the two ideas. Use a simple example.
22. How is ■ Vectors related to dot/cross product? Compare the two ideas. Use a school-life example.
23. How is dot/cross product related to geometric applications.? Compare the two ideas. Use a real-world example.
24. How is geometric applications. related to ■ Vectors? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Vectors related to dot/cross product? Compare the two ideas. Explain in your own words.
26. How is dot/cross product related to geometric applications.? Compare the two ideas. Give a step-by-step response.
27. How is geometric applications. related to ■ Vectors? Compare the two ideas. Mention one common misconception.
28. How is ■ Vectors related to dot/cross product? Compare the two ideas. State one practical application.
29. How is dot/cross product related to geometric applications.? Compare the two ideas. Justify your response.
30. How is geometric applications. related to ■ Vectors? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Vectors to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply dot/cross product to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply geometric applications. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Vectors to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply dot/cross product to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply geometric applications. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Vectors to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply dot/cross product to a realistic situation. What would you do or calculate? State one practical application.
39. Apply geometric applications. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Vectors to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of dot/cross product. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of geometric applications.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Vectors. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of dot/cross product. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of geometric applications.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Vectors. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of dot/cross product. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of geometric applications.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Vectors. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of dot/cross product. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show geometric applications.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Vectors. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show dot/cross product. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show geometric applications.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Vectors. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show dot/cross product. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show geometric applications.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Vectors. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show dot/cross product. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show geometric applications.. Include the relevant rule or principle.

MATHEMATICS — 11. Three-Dimensional Geometry

Coverage: ■ Direction cosines/ratios; line/plane equations; distances.

Understanding

1. What is ■ Direction cosines/ratios, and what are its main features? Use a simple example.
2. What is line/plane equations, and what are its main features? Use a school-life example.
3. What is distances., and what are its main features? Use a real-world example.
4. What is ■ Direction cosines/ratios, and what are its main features? Show the idea with a diagram where appropriate.
5. What is line/plane equations, and what are its main features? Explain in your own words.
6. What is distances., and what are its main features? Give a step-by-step response.
7. What is ■ Direction cosines/ratios, and what are its main features? Mention one common misconception.
8. What is line/plane equations, and what are its main features? State one practical application.
9. What is distances., and what are its main features? Justify your response.
10. What is ■ Direction cosines/ratios, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to line/plane equations and explain each. Use a simple example.
12. Give two examples related to distances. and explain each. Use a school-life example.
13. Give two examples related to ■ Direction cosines/ratios and explain each. Use a real-world example.
14. Give two examples related to line/plane equations and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to distances. and explain each. Explain in your own words.
16. Give two examples related to ■ Direction cosines/ratios and explain each. Give a step-by-step response.
17. Give two examples related to line/plane equations and explain each. Mention one common misconception.
18. Give two examples related to distances. and explain each. State one practical application.
19. Give two examples related to ■ Direction cosines/ratios and explain each. Justify your response.
20. Give two examples related to line/plane equations and explain each. Include the relevant rule or principle.

Comparison

21. How is distances. related to ■ Direction cosines/ratios? Compare the two ideas. Use a simple example.
22. How is ■ Direction cosines/ratios related to line/plane equations? Compare the two ideas. Use a school-life example.
23. How is line/plane equations related to distances.? Compare the two ideas. Use a real-world example.
24. How is distances. related to ■ Direction cosines/ratios? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Direction cosines/ratios related to line/plane equations? Compare the two ideas. Explain in your own words.
26. How is line/plane equations related to distances.? Compare the two ideas. Give a step-by-step response.
27. How is distances. related to ■ Direction cosines/ratios? Compare the two ideas. Mention one common misconception.
28. How is ■ Direction cosines/ratios related to line/plane equations? Compare the two ideas. State one practical application.
29. How is line/plane equations related to distances.? Compare the two ideas. Justify your response.
30. How is distances. related to ■ Direction cosines/ratios? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Direction cosines/ratios to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply line/plane equations to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply distances. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Direction cosines/ratios to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply line/plane equations to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply distances. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Direction cosines/ratios to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply line/plane equations to a realistic situation. What would you do or calculate? State one practical application.
39. Apply distances. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Direction cosines/ratios to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of line/plane equations. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of distances.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Direction cosines/ratios. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of line/plane equations. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of distances.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Direction cosines/ratios. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of line/plane equations. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of distances.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Direction cosines/ratios. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of line/plane equations. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show distances.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Direction cosines/ratios. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show line/plane equations. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show distances.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Direction cosines/ratios. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show line/plane equations. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show distances.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Direction cosines/ratios. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show line/plane equations. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show distances.. Include the relevant rule or principle.

MATHEMATICS — 12. Linear Programming

Coverage: ■ Formulation; feasible region; graphical optimisation.

Understanding

1. What is ■ Formulation, and what are its main features? Use a simple example.
2. What is feasible region, and what are its main features? Use a school-life example.
3. What is graphical optimisation., and what are its main features? Use a real-world example.
4. What is ■ Formulation, and what are its main features? Show the idea with a diagram where appropriate.
5. What is feasible region, and what are its main features? Explain in your own words.
6. What is graphical optimisation., and what are its main features? Give a step-by-step response.
7. What is ■ Formulation, and what are its main features? Mention one common misconception.
8. What is feasible region, and what are its main features? State one practical application.
9. What is graphical optimisation., and what are its main features? Justify your response.
10. What is ■ Formulation, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to feasible region and explain each. Use a simple example.
12. Give two examples related to graphical optimisation. and explain each. Use a school-life example.
13. Give two examples related to ■ Formulation and explain each. Use a real-world example.
14. Give two examples related to feasible region and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to graphical optimisation. and explain each. Explain in your own words.
16. Give two examples related to ■ Formulation and explain each. Give a step-by-step response.
17. Give two examples related to feasible region and explain each. Mention one common misconception.
18. Give two examples related to graphical optimisation. and explain each. State one practical application.
19. Give two examples related to ■ Formulation and explain each. Justify your response.
20. Give two examples related to feasible region and explain each. Include the relevant rule or principle.

Comparison

21. How is graphical optimisation. related to ■ Formulation? Compare the two ideas. Use a simple example.
22. How is ■ Formulation related to feasible region? Compare the two ideas. Use a school-life example.
23. How is feasible region related to graphical optimisation.? Compare the two ideas. Use a real-world example.
24. How is graphical optimisation. related to ■ Formulation? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Formulation related to feasible region? Compare the two ideas. Explain in your own words.
26. How is feasible region related to graphical optimisation.? Compare the two ideas. Give a step-by-step response.
27. How is graphical optimisation. related to ■ Formulation? Compare the two ideas. Mention one common misconception.
28. How is ■ Formulation related to feasible region? Compare the two ideas. State one practical application.
29. How is feasible region related to graphical optimisation.? Compare the two ideas. Justify your response.
30. How is graphical optimisation. related to ■ Formulation? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Formulation to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply feasible region to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply graphical optimisation. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Formulation to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply feasible region to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply graphical optimisation. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Formulation to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply feasible region to a realistic situation. What would you do or calculate? State one practical application.
39. Apply graphical optimisation. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Formulation to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of feasible region. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of graphical optimisation.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Formulation. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of feasible region. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of graphical optimisation.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Formulation. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of feasible region. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of graphical optimisation.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Formulation. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of feasible region. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show graphical optimisation.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Formulation. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show feasible region. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show graphical optimisation.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Formulation. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show feasible region. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show graphical optimisation.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Formulation. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show feasible region. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show graphical optimisation.. Include the relevant rule or principle.

MATHEMATICS — 13. Probability

Coverage: ■ Conditional probability; Bayes theorem; random variables; distributions.

Understanding

1. What is ■ Conditional probability, and what are its main features? Use a simple example.
2. What is Bayes theorem, and what are its main features? Use a school-life example.
3. What is random variables, and what are its main features? Use a real-world example.
4. What is distributions., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Conditional probability, and what are its main features? Explain in your own words.
6. What is Bayes theorem, and what are its main features? Give a step-by-step response.
7. What is random variables, and what are its main features? Mention one common misconception.
8. What is distributions., and what are its main features? State one practical application.
9. What is ■ Conditional probability, and what are its main features? Justify your response.
10. What is Bayes theorem, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to random variables and explain each. Use a simple example.
12. Give two examples related to distributions. and explain each. Use a school-life example.
13. Give two examples related to ■ Conditional probability and explain each. Use a real-world example.
14. Give two examples related to Bayes theorem and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to random variables and explain each. Explain in your own words.
16. Give two examples related to distributions. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Conditional probability and explain each. Mention one common misconception.
18. Give two examples related to Bayes theorem and explain each. State one practical application.
19. Give two examples related to random variables and explain each. Justify your response.
20. Give two examples related to distributions. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Conditional probability related to Bayes theorem? Compare the two ideas. Use a simple example.
22. How is Bayes theorem related to random variables? Compare the two ideas. Use a school-life example.
23. How is random variables related to distributions.? Compare the two ideas. Use a real-world example.
24. How is distributions. related to ■ Conditional probability? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Conditional probability related to Bayes theorem? Compare the two ideas. Explain in your own words.
26. How is Bayes theorem related to random variables? Compare the two ideas. Give a step-by-step response.
27. How is random variables related to distributions.? Compare the two ideas. Mention one common misconception.
28. How is distributions. related to ■ Conditional probability? Compare the two ideas. State one practical application.
29. How is ■ Conditional probability related to Bayes theorem? Compare the two ideas. Justify your response.
30. How is Bayes theorem related to random variables? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply random variables to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply distributions. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Conditional probability to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply Bayes theorem to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply random variables to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply distributions. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Conditional probability to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply Bayes theorem to a realistic situation. What would you do or calculate? State one practical application.
39. Apply random variables to a realistic situation. What would you do or calculate? Justify your response.
40. Apply distributions. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Conditional probability. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of Bayes theorem. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of random variables. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of distributions.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Conditional probability. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of Bayes theorem. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of random variables. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of distributions.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Conditional probability. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of Bayes theorem. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show random variables. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show distributions.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Conditional probability. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show Bayes theorem. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show random variables. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show distributions.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Conditional probability. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show Bayes theorem. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show random variables. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show distributions.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — Electrostatics

Coverage: ■ Electric charges/fields; Gauss law; potential; capacitance.

Understanding

1. What is ■ Electric charges/fields, and what are its main features? Use a simple example.
2. What is Gauss law, and what are its main features? Use a school-life example.
3. What is potential, and what are its main features? Use a real-world example.
4. What is capacitance., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Electric charges/fields, and what are its main features? Explain in your own words.
6. What is Gauss law, and what are its main features? Give a step-by-step response.
7. What is potential, and what are its main features? Mention one common misconception.
8. What is capacitance., and what are its main features? State one practical application.
9. What is ■ Electric charges/fields, and what are its main features? Justify your response.
10. What is Gauss law, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to potential and explain each. Use a simple example.
12. Give two examples related to capacitance. and explain each. Use a school-life example.
13. Give two examples related to ■ Electric charges/fields and explain each. Use a real-world example.
14. Give two examples related to Gauss law and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to potential and explain each. Explain in your own words.
16. Give two examples related to capacitance. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Electric charges/fields and explain each. Mention one common misconception.
18. Give two examples related to Gauss law and explain each. State one practical application.
19. Give two examples related to potential and explain each. Justify your response.
20. Give two examples related to capacitance. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Electric charges/fields related to Gauss law? Compare the two ideas. Use a simple example.
22. How is Gauss law related to potential? Compare the two ideas. Use a school-life example.
23. How is potential related to capacitance.? Compare the two ideas. Use a real-world example.
24. How is capacitance. related to ■ Electric charges/fields? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Electric charges/fields related to Gauss law? Compare the two ideas. Explain in your own words.
26. How is Gauss law related to potential? Compare the two ideas. Give a step-by-step response.
27. How is potential related to capacitance.? Compare the two ideas. Mention one common misconception.
28. How is capacitance. related to ■ Electric charges/fields? Compare the two ideas. State one practical application.
29. How is ■ Electric charges/fields related to Gauss law? Compare the two ideas. Justify your response.
30. How is Gauss law related to potential? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply potential to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply capacitance. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Electric charges/fields to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply Gauss law to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply potential to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply capacitance. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Electric charges/fields to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply Gauss law to a realistic situation. What would you do or calculate? State one practical application.
39. Apply potential to a realistic situation. What would you do or calculate? Justify your response.
40. Apply capacitance. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Electric charges/fields. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of Gauss law. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of potential. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of capacitance.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Electric charges/fields. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of Gauss law. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of potential. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of capacitance.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Electric charges/fields. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of Gauss law. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show potential. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show capacitance.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Electric charges/fields. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show Gauss law. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show potential. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show capacitance.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Electric charges/fields. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show Gauss law. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show potential. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show capacitance.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — Current Electricity

Coverage: ■ Ohm's law; resistivity; circuits; Kirchhoff laws.

Understanding

1. What is ■ Ohm's law, and what are its main features? Use a simple example.
2. What is resistivity, and what are its main features? Use a school-life example.
3. What is circuits, and what are its main features? Use a real-world example.
4. What is Kirchhoff laws., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Ohm's law, and what are its main features? Explain in your own words.
6. What is resistivity, and what are its main features? Give a step-by-step response.
7. What is circuits, and what are its main features? Mention one common misconception.
8. What is Kirchhoff laws., and what are its main features? State one practical application.
9. What is ■ Ohm's law, and what are its main features? Justify your response.
10. What is resistivity, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to circuits and explain each. Use a simple example.
12. Give two examples related to Kirchhoff laws. and explain each. Use a school-life example.
13. Give two examples related to ■ Ohm's law and explain each. Use a real-world example.
14. Give two examples related to resistivity and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to circuits and explain each. Explain in your own words.
16. Give two examples related to Kirchhoff laws. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Ohm's law and explain each. Mention one common misconception.
18. Give two examples related to resistivity and explain each. State one practical application.
19. Give two examples related to circuits and explain each. Justify your response.
20. Give two examples related to Kirchhoff laws. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Ohm's law related to resistivity? Compare the two ideas. Use a simple example.
22. How is resistivity related to circuits? Compare the two ideas. Use a school-life example.
23. How is circuits related to Kirchhoff laws.? Compare the two ideas. Use a real-world example.
24. How is Kirchhoff laws. related to ■ Ohm's law? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Ohm's law related to resistivity? Compare the two ideas. Explain in your own words.
26. How is resistivity related to circuits? Compare the two ideas. Give a step-by-step response.
27. How is circuits related to Kirchhoff laws.? Compare the two ideas. Mention one common misconception.
28. How is Kirchhoff laws. related to ■ Ohm's law? Compare the two ideas. State one practical application.
29. How is ■ Ohm's law related to resistivity? Compare the two ideas. Justify your response.
30. How is resistivity related to circuits? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply circuits to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply Kirchhoff laws. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Ohm's law to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply resistivity to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply circuits to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply Kirchhoff laws. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Ohm's law to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply resistivity to a realistic situation. What would you do or calculate? State one practical application.
39. Apply circuits to a realistic situation. What would you do or calculate? Justify your response.
40. Apply Kirchhoff laws. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Ohm's law. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of resistivity. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of circuits. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of Kirchhoff laws.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Ohm's law. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of resistivity. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of circuits. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of Kirchhoff laws.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Ohm's law. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of resistivity. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show circuits. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show Kirchhoff laws.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Ohm's law. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show resistivity. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show circuits. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show Kirchhoff laws.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Ohm's law. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show resistivity. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show circuits. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show Kirchhoff laws.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — Magnetism & Electromagnetic Induction

Coverage: ■ Moving charges; magnetic fields; Faraday/Lenz; inductance; AC.

Understanding

1. What is ■ Moving charges, and what are its main features? Use a simple example.
2. What is magnetic fields, and what are its main features? Use a school-life example.
3. What is Faraday/Lenz, and what are its main features? Use a real-world example.
4. What is inductance, and what are its main features? Show the idea with a diagram where appropriate.
5. What is AC., and what are its main features? Explain in your own words.
6. What is ■ Moving charges, and what are its main features? Give a step-by-step response.
7. What is magnetic fields, and what are its main features? Mention one common misconception.
8. What is Faraday/Lenz, and what are its main features? State one practical application.
9. What is inductance, and what are its main features? Justify your response.
10. What is AC., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Moving charges and explain each. Use a simple example.
12. Give two examples related to magnetic fields and explain each. Use a school-life example.
13. Give two examples related to Faraday/Lenz and explain each. Use a real-world example.
14. Give two examples related to inductance and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to AC. and explain each. Explain in your own words.
16. Give two examples related to ■ Moving charges and explain each. Give a step-by-step response.
17. Give two examples related to magnetic fields and explain each. Mention one common misconception.
18. Give two examples related to Faraday/Lenz and explain each. State one practical application.
19. Give two examples related to inductance and explain each. Justify your response.
20. Give two examples related to AC. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Moving charges related to magnetic fields? Compare the two ideas. Use a simple example.
22. How is magnetic fields related to Faraday/Lenz? Compare the two ideas. Use a school-life example.
23. How is Faraday/Lenz related to inductance? Compare the two ideas. Use a real-world example.
24. How is inductance related to AC.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is AC. related to ■ Moving charges? Compare the two ideas. Explain in your own words.
26. How is ■ Moving charges related to magnetic fields? Compare the two ideas. Give a step-by-step response.
27. How is magnetic fields related to Faraday/Lenz? Compare the two ideas. Mention one common misconception.
28. How is Faraday/Lenz related to inductance? Compare the two ideas. State one practical application.
29. How is inductance related to AC.? Compare the two ideas. Justify your response.
30. How is AC. related to ■ Moving charges? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Moving charges to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply magnetic fields to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply Faraday/Lenz to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply inductance to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply AC. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Moving charges to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply magnetic fields to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply Faraday/Lenz to a realistic situation. What would you do or calculate? State one practical application.
39. Apply inductance to a realistic situation. What would you do or calculate? Justify your response.
40. Apply AC. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Moving charges. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of magnetic fields. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of Faraday/Lenz. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of inductance. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of AC.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Moving charges. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of magnetic fields. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of Faraday/Lenz. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of inductance. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of AC.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Moving charges. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show magnetic fields. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show Faraday/Lenz. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show inductance. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show AC.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Moving charges. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show magnetic fields. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show Faraday/Lenz. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show inductance. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show AC.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — Electromagnetic Waves

Coverage: ■ Spectrum; properties; applications.

Understanding

1. What is ■ Spectrum, and what are its main features? Use a simple example.
2. What is properties, and what are its main features? Use a school-life example.
3. What is applications., and what are its main features? Use a real-world example.
4. What is ■ Spectrum, and what are its main features? Show the idea with a diagram where appropriate.
5. What is properties, and what are its main features? Explain in your own words.
6. What is applications., and what are its main features? Give a step-by-step response.
7. What is ■ Spectrum, and what are its main features? Mention one common misconception.
8. What is properties, and what are its main features? State one practical application.
9. What is applications., and what are its main features? Justify your response.
10. What is ■ Spectrum, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to properties and explain each. Use a simple example.
12. Give two examples related to applications. and explain each. Use a school-life example.
13. Give two examples related to ■ Spectrum and explain each. Use a real-world example.
14. Give two examples related to properties and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to applications. and explain each. Explain in your own words.
16. Give two examples related to ■ Spectrum and explain each. Give a step-by-step response.
17. Give two examples related to properties and explain each. Mention one common misconception.
18. Give two examples related to applications. and explain each. State one practical application.
19. Give two examples related to ■ Spectrum and explain each. Justify your response.
20. Give two examples related to properties and explain each. Include the relevant rule or principle.

Comparison

21. How is applications. related to ■ Spectrum? Compare the two ideas. Use a simple example.
22. How is ■ Spectrum related to properties? Compare the two ideas. Use a school-life example.
23. How is properties related to applications.? Compare the two ideas. Use a real-world example.
24. How is applications. related to ■ Spectrum? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Spectrum related to properties? Compare the two ideas. Explain in your own words.
26. How is properties related to applications.? Compare the two ideas. Give a step-by-step response.
27. How is applications. related to ■ Spectrum? Compare the two ideas. Mention one common misconception.
28. How is ■ Spectrum related to properties? Compare the two ideas. State one practical application.
29. How is properties related to applications.? Compare the two ideas. Justify your response.
30. How is applications. related to ■ Spectrum? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Spectrum to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply properties to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply applications. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Spectrum to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply properties to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply applications. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Spectrum to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply properties to a realistic situation. What would you do or calculate? State one practical application.
39. Apply applications. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Spectrum to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of properties. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Spectrum. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of properties. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Spectrum. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of properties. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Spectrum. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of properties. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Spectrum. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show properties. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Spectrum. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show properties. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Spectrum. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show properties. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — Optics

Coverage: NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus

Understanding

1. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a simple example.
2. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a school-life example.
3. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a real-world example.
4. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Show the idea with a diagram where appropriate.
5. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Explain in your own words.
6. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Give a step-by-step response.
7. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Mention one common misconception.
8. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? State one practical application.
9. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Justify your response.
10. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a simple example.
12. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a school-life example.
13. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a real-world example.
14. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Show the idea with a diagram where appropriate.
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16. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Give a step-by-step response.
17. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Mention one common misconception.
18. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. State one practical application.
19. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Justify your response.
20. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Include the relevant rule or principle.

Comparison

21. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a simple example.
22. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a school-life example.
23. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a real-world example.

24. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Explain in your own words.
26. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Give a step-by-step response.
27. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Mention one common misconception.
28. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. State one practical application.
29. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Justify your response.
30. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? State one practical application.
39. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Justify your response.
40. Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a real-world example.
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48. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. State one practical application.

49. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Justify your response.

50. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a simple example.

52. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a school-life example.

53. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a real-world example.

54. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Show the idea with a diagram where appropriate.

55. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Explain in your own words.

56. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Give a step-by-step response.

57. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Mention one common misconception.

58. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — Dual Nature, Atoms & Nuclei

Coverage: ■ Photoelectric effect; de Broglie; Bohr model; spectra; radioactivity; binding energy.

Understanding

1. What is ■ Photoelectric effect, and what are its main features? Use a simple example.
2. What is de Broglie, and what are its main features? Use a school-life example.
3. What is Bohr model, and what are its main features? Use a real-world example.
4. What is spectra, and what are its main features? Show the idea with a diagram where appropriate.
5. What is radioactivity, and what are its main features? Explain in your own words.
6. What is binding energy, and what are its main features? Give a step-by-step response.
7. What is ■ Photoelectric effect, and what are its main features? Mention one common misconception.
8. What is de Broglie, and what are its main features? State one practical application.
9. What is Bohr model, and what are its main features? Justify your response.
10. What is spectra, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to radioactivity and explain each. Use a simple example.
12. Give two examples related to binding energy, and explain each. Use a school-life example.
13. Give two examples related to ■ Photoelectric effect and explain each. Use a real-world example.
14. Give two examples related to de Broglie and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to Bohr model and explain each. Explain in your own words.
16. Give two examples related to spectra and explain each. Give a step-by-step response.
17. Give two examples related to radioactivity and explain each. Mention one common misconception.
18. Give two examples related to binding energy, and explain each. State one practical application.
19. Give two examples related to ■ Photoelectric effect and explain each. Justify your response.
20. Give two examples related to de Broglie and explain each. Include the relevant rule or principle.

Comparison

21. How is Bohr model related to spectra? Compare the two ideas. Use a simple example.
22. How is spectra related to radioactivity? Compare the two ideas. Use a school-life example.
23. How is radioactivity related to binding energy? Compare the two ideas. Use a real-world example.
24. How is binding energy, related to ■ Photoelectric effect? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Photoelectric effect related to de Broglie? Compare the two ideas. Explain in your own words.
26. How is de Broglie related to Bohr model? Compare the two ideas. Give a step-by-step response.
27. How is Bohr model related to spectra? Compare the two ideas. Mention one common misconception.
28. How is spectra related to radioactivity? Compare the two ideas. State one practical application.
29. How is radioactivity related to binding energy? Compare the two ideas. Justify your response.
30. How is binding energy, related to ■ Photoelectric effect? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Photoelectric effect to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply de Broglie to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply Bohr model to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply spectra to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply radioactivity to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply binding energy. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Photoelectric effect to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply de Broglie to a realistic situation. What would you do or calculate? State one practical application.
39. Apply Bohr model to a realistic situation. What would you do or calculate? Justify your response.
40. Apply spectra to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of radioactivity. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of binding energy.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Photoelectric effect. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of de Broglie. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of Bohr model. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of spectra. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of radioactivity. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of binding energy.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Photoelectric effect. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of de Broglie. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show Bohr model. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show spectra. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show radioactivity. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show binding energy.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Photoelectric effect. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show de Broglie. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show Bohr model. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show spectra. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show radioactivity. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show binding energy.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — PHYSICS — Semiconductor Electronics

Coverage: ■ Diodes; transistors; logic gates; electronic applications.

Understanding

1. What is ■ Diodes, and what are its main features? Use a simple example.
2. What is transistors, and what are its main features? Use a school-life example.
3. What is logic gates, and what are its main features? Use a real-world example.
4. What is electronic applications., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Diodes, and what are its main features? Explain in your own words.
6. What is transistors, and what are its main features? Give a step-by-step response.
7. What is logic gates, and what are its main features? Mention one common misconception.
8. What is electronic applications., and what are its main features? State one practical application.
9. What is ■ Diodes, and what are its main features? Justify your response.
10. What is transistors, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to logic gates and explain each. Use a simple example.
12. Give two examples related to electronic applications. and explain each. Use a school-life example.
13. Give two examples related to ■ Diodes and explain each. Use a real-world example.
14. Give two examples related to transistors and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to logic gates and explain each. Explain in your own words.
16. Give two examples related to electronic applications. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Diodes and explain each. Mention one common misconception.
18. Give two examples related to transistors and explain each. State one practical application.
19. Give two examples related to logic gates and explain each. Justify your response.
20. Give two examples related to electronic applications. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Diodes related to transistors? Compare the two ideas. Use a simple example.
22. How is transistors related to logic gates? Compare the two ideas. Use a school-life example.
23. How is logic gates related to electronic applications.? Compare the two ideas. Use a real-world example.
24. How is electronic applications. related to ■ Diodes? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Diodes related to transistors? Compare the two ideas. Explain in your own words.
26. How is transistors related to logic gates? Compare the two ideas. Give a step-by-step response.
27. How is logic gates related to electronic applications.? Compare the two ideas. Mention one common misconception.
28. How is electronic applications. related to ■ Diodes? Compare the two ideas. State one practical application.
29. How is ■ Diodes related to transistors? Compare the two ideas. Justify your response.
30. How is transistors related to logic gates? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply logic gates to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply electronic applications. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Diodes to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply transistors to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply logic gates to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply electronic applications. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Diodes to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply transistors to a realistic situation. What would you do or calculate? State one practical application.
39. Apply logic gates to a realistic situation. What would you do or calculate? Justify your response.
40. Apply electronic applications. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Diodes. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of transistors. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of logic gates. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of electronic applications.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Diodes. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of transistors. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of logic gates. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of electronic applications.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Diodes. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of transistors. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show logic gates. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show electronic applications.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Diodes. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show transistors. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show logic gates. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show electronic applications.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Diodes. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show transistors. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show logic gates. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show electronic applications.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — CHEMISTRY — Solutions

Coverage: ■ Concentration; colligative properties; Raoult's law.

Understanding

1. What is ■ Concentration, and what are its main features? Use a simple example.
2. What is colligative properties, and what are its main features? Use a school-life example.
3. What is Raoult's law., and what are its main features? Use a real-world example.
4. What is ■ Concentration, and what are its main features? Show the idea with a diagram where appropriate.
5. What is colligative properties, and what are its main features? Explain in your own words.
6. What is Raoult's law., and what are its main features? Give a step-by-step response.
7. What is ■ Concentration, and what are its main features? Mention one common misconception.
8. What is colligative properties, and what are its main features? State one practical application.
9. What is Raoult's law., and what are its main features? Justify your response.
10. What is ■ Concentration, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to colligative properties and explain each. Use a simple example.
12. Give two examples related to Raoult's law. and explain each. Use a school-life example.
13. Give two examples related to ■ Concentration and explain each. Use a real-world example.
14. Give two examples related to colligative properties and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to Raoult's law. and explain each. Explain in your own words.
16. Give two examples related to ■ Concentration and explain each. Give a step-by-step response.
17. Give two examples related to colligative properties and explain each. Mention one common misconception.
18. Give two examples related to Raoult's law. and explain each. State one practical application.
19. Give two examples related to ■ Concentration and explain each. Justify your response.
20. Give two examples related to colligative properties and explain each. Include the relevant rule or principle.

Comparison

21. How is Raoult's law. related to ■ Concentration? Compare the two ideas. Use a simple example.
22. How is ■ Concentration related to colligative properties? Compare the two ideas. Use a school-life example.
23. How is colligative properties related to Raoult's law.? Compare the two ideas. Use a real-world example.
24. How is Raoult's law. related to ■ Concentration? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Concentration related to colligative properties? Compare the two ideas. Explain in your own words.
26. How is colligative properties related to Raoult's law.? Compare the two ideas. Give a step-by-step response.
27. How is Raoult's law. related to ■ Concentration? Compare the two ideas. Mention one common misconception.
28. How is ■ Concentration related to colligative properties? Compare the two ideas. State one practical application.
29. How is colligative properties related to Raoult's law.? Compare the two ideas. Justify your response.
30. How is Raoult's law. related to ■ Concentration? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Concentration to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply colligative properties to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply Raoult's law. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Concentration to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply colligative properties to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply Raoult's law. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Concentration to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply colligative properties to a realistic situation. What would you do or calculate? State one practical application.
39. Apply Raoult's law. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Concentration to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of colligative properties. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of Raoult's law.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Concentration. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of colligative properties. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of Raoult's law.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Concentration. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of colligative properties. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of Raoult's law.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Concentration. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of colligative properties. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show Raoult's law.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Concentration. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show colligative properties. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show Raoult's law.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Concentration. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show colligative properties. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show Raoult's law.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Concentration. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show colligative properties. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show Raoult's law.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — CHEMISTRY —

Electrochemistry

Coverage: ■ Cells; electrode potential; Nernst equation; conductance.

Understanding

1. What is ■ Cells, and what are its main features? Use a simple example.
2. What is electrode potential, and what are its main features? Use a school-life example.
3. What is Nernst equation, and what are its main features? Use a real-world example.
4. What is conductance., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Cells, and what are its main features? Explain in your own words.
6. What is electrode potential, and what are its main features? Give a step-by-step response.
7. What is Nernst equation, and what are its main features? Mention one common misconception.
8. What is conductance., and what are its main features? State one practical application.
9. What is ■ Cells, and what are its main features? Justify your response.
10. What is electrode potential, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to Nernst equation and explain each. Use a simple example.
12. Give two examples related to conductance. and explain each. Use a school-life example.
13. Give two examples related to ■ Cells and explain each. Use a real-world example.
14. Give two examples related to electrode potential and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to Nernst equation and explain each. Explain in your own words.
16. Give two examples related to conductance. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Cells and explain each. Mention one common misconception.
18. Give two examples related to electrode potential and explain each. State one practical application.
19. Give two examples related to Nernst equation and explain each. Justify your response.
20. Give two examples related to conductance. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Cells related to electrode potential? Compare the two ideas. Use a simple example.
22. How is electrode potential related to Nernst equation? Compare the two ideas. Use a school-life example.
23. How is Nernst equation related to conductance.? Compare the two ideas. Use a real-world example.
24. How is conductance. related to ■ Cells? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Cells related to electrode potential? Compare the two ideas. Explain in your own words.
26. How is electrode potential related to Nernst equation? Compare the two ideas. Give a step-by-step response.
27. How is Nernst equation related to conductance.? Compare the two ideas. Mention one common misconception.
28. How is conductance. related to ■ Cells? Compare the two ideas. State one practical application.
29. How is ■ Cells related to electrode potential? Compare the two ideas. Justify your response.
30. How is electrode potential related to Nernst equation? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply Nernst equation to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply conductance. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Cells to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply electrode potential to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply Nernst equation to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply conductance. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Cells to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply electrode potential to a realistic situation. What would you do or calculate? State one practical application.
39. Apply Nernst equation to a realistic situation. What would you do or calculate? Justify your response.
40. Apply conductance. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Cells. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of electrode potential. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of Nernst equation. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of conductance.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Cells. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of electrode potential. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of Nernst equation. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of conductance.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Cells. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of electrode potential. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show Nernst equation. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show conductance.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Cells. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show electrode potential. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show Nernst equation. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show conductance.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Cells. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show electrode potential. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show Nernst equation. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show conductance.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — CHEMISTRY — Chemical Kinetics

Coverage: ■ Rate; order; molecularity; Arrhenius equation.

Understanding

1. What is ■ Rate, and what are its main features? Use a simple example.
2. What is order, and what are its main features? Use a school-life example.
3. What is molecularity, and what are its main features? Use a real-world example.
4. What is Arrhenius equation., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Rate, and what are its main features? Explain in your own words.
6. What is order, and what are its main features? Give a step-by-step response.
7. What is molecularity, and what are its main features? Mention one common misconception.
8. What is Arrhenius equation., and what are its main features? State one practical application.
9. What is ■ Rate, and what are its main features? Justify your response.
10. What is order, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to molecularity and explain each. Use a simple example.
12. Give two examples related to Arrhenius equation. and explain each. Use a school-life example.
13. Give two examples related to ■ Rate and explain each. Use a real-world example.
14. Give two examples related to order and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to molecularity and explain each. Explain in your own words.
16. Give two examples related to Arrhenius equation. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Rate and explain each. Mention one common misconception.
18. Give two examples related to order and explain each. State one practical application.
19. Give two examples related to molecularity and explain each. Justify your response.
20. Give two examples related to Arrhenius equation. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Rate related to order? Compare the two ideas. Use a simple example.
22. How is order related to molecularity? Compare the two ideas. Use a school-life example.
23. How is molecularity related to Arrhenius equation.? Compare the two ideas. Use a real-world example.
24. How is Arrhenius equation. related to ■ Rate? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Rate related to order? Compare the two ideas. Explain in your own words.
26. How is order related to molecularity? Compare the two ideas. Give a step-by-step response.
27. How is molecularity related to Arrhenius equation.? Compare the two ideas. Mention one common misconception.
28. How is Arrhenius equation. related to ■ Rate? Compare the two ideas. State one practical application.
29. How is ■ Rate related to order? Compare the two ideas. Justify your response.
30. How is order related to molecularity? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply molecularity to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply Arrhenius equation. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Rate to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply order to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply molecularity to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply Arrhenius equation. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Rate to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply order to a realistic situation. What would you do or calculate? State one practical application.
39. Apply molecularity to a realistic situation. What would you do or calculate? Justify your response.
40. Apply Arrhenius equation. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Rate. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of order. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of molecularity. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of Arrhenius equation.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Rate. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of order. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of molecularity. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of Arrhenius equation.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Rate. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of order. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show molecularity. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show Arrhenius equation.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Rate. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show order. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show molecularity. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show Arrhenius equation.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Rate. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show order. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show molecularity. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show Arrhenius equation.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — CHEMISTRY — d/f Block & Coordination Compounds

Coverage: ■ Transition elements; coordination nomenclature; bonding; isomerism.

Understanding

1. What is ■ Transition elements, and what are its main features? Use a simple example.
2. What is coordination nomenclature, and what are its main features? Use a school-life example.
3. What is bonding, and what are its main features? Use a real-world example.
4. What is isomerism., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Transition elements, and what are its main features? Explain in your own words.
6. What is coordination nomenclature, and what are its main features? Give a step-by-step response.
7. What is bonding, and what are its main features? Mention one common misconception.
8. What is isomerism., and what are its main features? State one practical application.
9. What is ■ Transition elements, and what are its main features? Justify your response.
10. What is coordination nomenclature, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to bonding and explain each. Use a simple example.
12. Give two examples related to isomerism. and explain each. Use a school-life example.
13. Give two examples related to ■ Transition elements and explain each. Use a real-world example.
14. Give two examples related to coordination nomenclature and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to bonding and explain each. Explain in your own words.
16. Give two examples related to isomerism. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Transition elements and explain each. Mention one common misconception.
18. Give two examples related to coordination nomenclature and explain each. State one practical application.
19. Give two examples related to bonding and explain each. Justify your response.
20. Give two examples related to isomerism. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Transition elements related to coordination nomenclature? Compare the two ideas. Use a simple example.
22. How is coordination nomenclature related to bonding? Compare the two ideas. Use a school-life example.
23. How is bonding related to isomerism.? Compare the two ideas. Use a real-world example.
24. How is isomerism. related to ■ Transition elements? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Transition elements related to coordination nomenclature? Compare the two ideas. Explain in your own words.
26. How is coordination nomenclature related to bonding? Compare the two ideas. Give a step-by-step response.
27. How is bonding related to isomerism.? Compare the two ideas. Mention one common misconception.
28. How is isomerism. related to ■ Transition elements? Compare the two ideas. State one practical application.
29. How is ■ Transition elements related to coordination nomenclature? Compare the two ideas. Justify your response.
30. How is coordination nomenclature related to bonding? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply bonding to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply isomerism. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Transition elements to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply coordination nomenclature to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply bonding to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply isomerism. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Transition elements to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply coordination nomenclature to a realistic situation. What would you do or calculate? State one practical application.
39. Apply bonding to a realistic situation. What would you do or calculate? Justify your response.
40. Apply isomerism. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Transition elements. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of coordination nomenclature. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of bonding. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of isomerism.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Transition elements. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of coordination nomenclature. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of bonding. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of isomerism.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Transition elements. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of coordination nomenclature. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show bonding. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show isomerism.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Transition elements. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show coordination nomenclature. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show bonding. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show isomerism.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Transition elements. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show coordination nomenclature. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show bonding. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show isomerism.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — CHEMISTRY — Organic Chemistry

Coverage: ■ Haloalkanes/haloarenes; alcohols/phenols/ethers; aldehydes/ketones/acids; amines.

Understanding

1. What is ■ Haloalkanes/haloarenes, and what are its main features? Use a simple example.
2. What is alcohols/phenols/ethers, and what are its main features? Use a school-life example.
3. What is aldehydes/ketones/acids, and what are its main features? Use a real-world example.
4. What is amines., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Haloalkanes/haloarenes, and what are its main features? Explain in your own words.
6. What is alcohols/phenols/ethers, and what are its main features? Give a step-by-step response.
7. What is aldehydes/ketones/acids, and what are its main features? Mention one common misconception.
8. What is amines., and what are its main features? State one practical application.
9. What is ■ Haloalkanes/haloarenes, and what are its main features? Justify your response.
10. What is alcohols/phenols/ethers, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to aldehydes/ketones/acids and explain each. Use a simple example.
12. Give two examples related to amines. and explain each. Use a school-life example.
13. Give two examples related to ■ Haloalkanes/haloarenes and explain each. Use a real-world example.
14. Give two examples related to alcohols/phenols/ethers and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to aldehydes/ketones/acids and explain each. Explain in your own words.
16. Give two examples related to amines. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Haloalkanes/haloarenes and explain each. Mention one common misconception.
18. Give two examples related to alcohols/phenols/ethers and explain each. State one practical application.
19. Give two examples related to aldehydes/ketones/acids and explain each. Justify your response.
20. Give two examples related to amines. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Haloalkanes/haloarenes related to alcohols/phenols/ethers? Compare the two ideas. Use a simple example.
22. How is alcohols/phenols/ethers related to aldehydes/ketones/acids? Compare the two ideas. Use a school-life example.
23. How is aldehydes/ketones/acids related to amines.? Compare the two ideas. Use a real-world example.
24. How is amines. related to ■ Haloalkanes/haloarenes? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Haloalkanes/haloarenes related to alcohols/phenols/ethers? Compare the two ideas. Explain in your own words.
26. How is alcohols/phenols/ethers related to aldehydes/ketones/acids? Compare the two ideas. Give a step-by-step response.
27. How is aldehydes/ketones/acids related to amines.? Compare the two ideas. Mention one common misconception.
28. How is amines. related to ■ Haloalkanes/haloarenes? Compare the two ideas. State one practical application.
29. How is ■ Haloalkanes/haloarenes related to alcohols/phenols/ethers? Compare the two ideas. Justify your response.
30. How is alcohols/phenols/ethers related to aldehydes/ketones/acids? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply aldehydes/ketones/acids to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply amines. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Haloalkanes/haloarenes to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply alcohols/phenols/ethers to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply aldehydes/ketones/acids to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply amines. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Haloalkanes/haloarenes to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply alcohols/phenols/ethers to a realistic situation. What would you do or calculate? State one practical application.
39. Apply aldehydes/ketones/acids to a realistic situation. What would you do or calculate? Justify your response.
40. Apply amines. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Haloalkanes/haloarenes. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of alcohols/phenols/ethers. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of aldehydes/ketones/acids. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of amines.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Haloalkanes/haloarenes. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of alcohols/phenols/ethers. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of aldehydes/ketones/acids. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of amines.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Haloalkanes/haloarenes. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of alcohols/phenols/ethers. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show aldehydes/ketones/acids. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show amines.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Haloalkanes/haloarenes. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show alcohols/phenols/ethers. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show aldehydes/ketones/acids. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show amines.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Haloalkanes/haloarenes. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show alcohols/phenols/ethers. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show aldehydes/ketones/acids. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show amines.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — CHEMISTRY —

Biomolecules

Coverage: ■ Carbohydrates; proteins; vitamins; nucleic acids.

Understanding

1. What is ■ Carbohydrates, and what are its main features? Use a simple example.
2. What is proteins, and what are its main features? Use a school-life example.
3. What is vitamins, and what are its main features? Use a real-world example.
4. What is nucleic acids., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Carbohydrates, and what are its main features? Explain in your own words.
6. What is proteins, and what are its main features? Give a step-by-step response.
7. What is vitamins, and what are its main features? Mention one common misconception.
8. What is nucleic acids., and what are its main features? State one practical application.
9. What is ■ Carbohydrates, and what are its main features? Justify your response.
10. What is proteins, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to vitamins and explain each. Use a simple example.
12. Give two examples related to nucleic acids. and explain each. Use a school-life example.
13. Give two examples related to ■ Carbohydrates and explain each. Use a real-world example.
14. Give two examples related to proteins and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to vitamins and explain each. Explain in your own words.
16. Give two examples related to nucleic acids. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Carbohydrates and explain each. Mention one common misconception.
18. Give two examples related to proteins and explain each. State one practical application.
19. Give two examples related to vitamins and explain each. Justify your response.
20. Give two examples related to nucleic acids. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Carbohydrates related to proteins? Compare the two ideas. Use a simple example.
22. How is proteins related to vitamins? Compare the two ideas. Use a school-life example.
23. How is vitamins related to nucleic acids.? Compare the two ideas. Use a real-world example.
24. How is nucleic acids. related to ■ Carbohydrates? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Carbohydrates related to proteins? Compare the two ideas. Explain in your own words.
26. How is proteins related to vitamins? Compare the two ideas. Give a step-by-step response.
27. How is vitamins related to nucleic acids.? Compare the two ideas. Mention one common misconception.
28. How is nucleic acids. related to ■ Carbohydrates? Compare the two ideas. State one practical application.
29. How is ■ Carbohydrates related to proteins? Compare the two ideas. Justify your response.
30. How is proteins related to vitamins? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply vitamins to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply nucleic acids. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Carbohydrates to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply proteins to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply vitamins to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply nucleic acids. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Carbohydrates to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply proteins to a realistic situation. What would you do or calculate? State one practical application.
39. Apply vitamins to a realistic situation. What would you do or calculate? Justify your response.
40. Apply nucleic acids. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Carbohydrates. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of proteins. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of vitamins. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of nucleic acids.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Carbohydrates. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of proteins. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of vitamins. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of nucleic acids.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Carbohydrates. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of proteins. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show vitamins. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show nucleic acids.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Carbohydrates. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show proteins. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show vitamins. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show nucleic acids.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Carbohydrates. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show proteins. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show vitamins. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show nucleic acids.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — BIOLOGY — Reproduction

Coverage: ■ Flowering plant reproduction; human reproduction; reproductive health.

Understanding

1. What is ■ Flowering plant reproduction, and what are its main features? Use a simple example.
2. What is human reproduction, and what are its main features? Use a school-life example.
3. What is reproductive health., and what are its main features? Use a real-world example.
4. What is ■ Flowering plant reproduction, and what are its main features? Show the idea with a diagram where appropriate.
5. What is human reproduction, and what are its main features? Explain in your own words.
6. What is reproductive health., and what are its main features? Give a step-by-step response.
7. What is ■ Flowering plant reproduction, and what are its main features? Mention one common misconception.
8. What is human reproduction, and what are its main features? State one practical application.
9. What is reproductive health., and what are its main features? Justify your response.
10. What is ■ Flowering plant reproduction, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to human reproduction and explain each. Use a simple example.
12. Give two examples related to reproductive health. and explain each. Use a school-life example.
13. Give two examples related to ■ Flowering plant reproduction and explain each. Use a real-world example.
14. Give two examples related to human reproduction and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to reproductive health. and explain each. Explain in your own words.
16. Give two examples related to ■ Flowering plant reproduction and explain each. Give a step-by-step response.
17. Give two examples related to human reproduction and explain each. Mention one common misconception.
18. Give two examples related to reproductive health. and explain each. State one practical application.
19. Give two examples related to ■ Flowering plant reproduction and explain each. Justify your response.
20. Give two examples related to human reproduction and explain each. Include the relevant rule or principle.

Comparison

21. How is reproductive health. related to ■ Flowering plant reproduction? Compare the two ideas. Use a simple example.
22. How is ■ Flowering plant reproduction related to human reproduction? Compare the two ideas. Use a school-life example.
23. How is human reproduction related to reproductive health.? Compare the two ideas. Use a real-world example.
24. How is reproductive health. related to ■ Flowering plant reproduction? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Flowering plant reproduction related to human reproduction? Compare the two ideas. Explain in your own words.
26. How is human reproduction related to reproductive health.? Compare the two ideas. Give a step-by-step response.
27. How is reproductive health. related to ■ Flowering plant reproduction? Compare the two ideas. Mention one common misconception.
28. How is ■ Flowering plant reproduction related to human reproduction? Compare the two ideas. State one practical application.
29. How is human reproduction related to reproductive health.? Compare the two ideas. Justify your response.
30. How is reproductive health. related to ■ Flowering plant reproduction? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Flowering plant reproduction to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply human reproduction to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply reproductive health. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Flowering plant reproduction to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply human reproduction to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply reproductive health. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Flowering plant reproduction to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply human reproduction to a realistic situation. What would you do or calculate? State one practical application.
39. Apply reproductive health. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Flowering plant reproduction to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of human reproduction. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of reproductive health.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Flowering plant reproduction. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of human reproduction. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of reproductive health.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Flowering plant reproduction. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of human reproduction. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of reproductive health.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Flowering plant reproduction. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of human reproduction. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show reproductive health.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Flowering plant reproduction. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show human reproduction. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show reproductive health.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Flowering plant reproduction. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show human reproduction. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show reproductive health.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Flowering plant reproduction. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show human reproduction. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show reproductive health.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — BIOLOGY — Genetics & Evolution

Coverage: ■ Inheritance; molecular basis; DNA/RNA; evolution.

Understanding

1. What is ■ Inheritance, and what are its main features? Use a simple example.
2. What is molecular basis, and what are its main features? Use a school-life example.
3. What is DNA/RNA, and what are its main features? Use a real-world example.
4. What is evolution., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Inheritance, and what are its main features? Explain in your own words.
6. What is molecular basis, and what are its main features? Give a step-by-step response.
7. What is DNA/RNA, and what are its main features? Mention one common misconception.
8. What is evolution., and what are its main features? State one practical application.
9. What is ■ Inheritance, and what are its main features? Justify your response.
10. What is molecular basis, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to DNA/RNA and explain each. Use a simple example.
12. Give two examples related to evolution. and explain each. Use a school-life example.
13. Give two examples related to ■ Inheritance and explain each. Use a real-world example.
14. Give two examples related to molecular basis and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to DNA/RNA and explain each. Explain in your own words.
16. Give two examples related to evolution. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Inheritance and explain each. Mention one common misconception.
18. Give two examples related to molecular basis and explain each. State one practical application.
19. Give two examples related to DNA/RNA and explain each. Justify your response.
20. Give two examples related to evolution. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Inheritance related to molecular basis? Compare the two ideas. Use a simple example.
22. How is molecular basis related to DNA/RNA? Compare the two ideas. Use a school-life example.
23. How is DNA/RNA related to evolution.? Compare the two ideas. Use a real-world example.
24. How is evolution. related to ■ Inheritance? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Inheritance related to molecular basis? Compare the two ideas. Explain in your own words.
26. How is molecular basis related to DNA/RNA? Compare the two ideas. Give a step-by-step response.
27. How is DNA/RNA related to evolution.? Compare the two ideas. Mention one common misconception.
28. How is evolution. related to ■ Inheritance? Compare the two ideas. State one practical application.
29. How is ■ Inheritance related to molecular basis? Compare the two ideas. Justify your response.
30. How is molecular basis related to DNA/RNA? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply DNA/RNA to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply evolution. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Inheritance to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply molecular basis to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply DNA/RNA to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply evolution. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Inheritance to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply molecular basis to a realistic situation. What would you do or calculate? State one practical application.
39. Apply DNA/RNA to a realistic situation. What would you do or calculate? Justify your response.
40. Apply evolution. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Inheritance. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of molecular basis. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of DNA/RNA. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of evolution.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Inheritance. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of molecular basis. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of DNA/RNA. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of evolution.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Inheritance. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of molecular basis. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show DNA/RNA. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show evolution.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Inheritance. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show molecular basis. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show DNA/RNA. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show evolution.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Inheritance. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show molecular basis. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show DNA/RNA. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show evolution.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — BIOLOGY — Human Welfare

Coverage: ■ Human health/disease; microbes and welfare.

Understanding

1. What is ■ Human health/disease, and what are its main features? Use a simple example.
2. What is microbes and welfare., and what are its main features? Use a school-life example.
3. What is ■ Human health/disease, and what are its main features? Use a real-world example.
4. What is microbes and welfare., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Human health/disease, and what are its main features? Explain in your own words.
6. What is microbes and welfare., and what are its main features? Give a step-by-step response.
7. What is ■ Human health/disease, and what are its main features? Mention one common misconception.
8. What is microbes and welfare., and what are its main features? State one practical application.
9. What is ■ Human health/disease, and what are its main features? Justify your response.
10. What is microbes and welfare., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Human health/disease and explain each. Use a simple example.
12. Give two examples related to microbes and welfare. and explain each. Use a school-life example.
13. Give two examples related to ■ Human health/disease and explain each. Use a real-world example.
14. Give two examples related to microbes and welfare. and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to ■ Human health/disease and explain each. Explain in your own words.
16. Give two examples related to microbes and welfare. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Human health/disease and explain each. Mention one common misconception.
18. Give two examples related to microbes and welfare. and explain each. State one practical application.
19. Give two examples related to ■ Human health/disease and explain each. Justify your response.
20. Give two examples related to microbes and welfare. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Human health/disease related to microbes and welfare.? Compare the two ideas. Use a simple example.
22. How is microbes and welfare. related to ■ Human health/disease? Compare the two ideas. Use a school-life example.
23. How is ■ Human health/disease related to microbes and welfare.? Compare the two ideas. Use a real-world example.
24. How is microbes and welfare. related to ■ Human health/disease? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Human health/disease related to microbes and welfare.? Compare the two ideas. Explain in your own words.
26. How is microbes and welfare. related to ■ Human health/disease? Compare the two ideas. Give a step-by-step response.
27. How is ■ Human health/disease related to microbes and welfare.? Compare the two ideas. Mention one common misconception.
28. How is microbes and welfare. related to ■ Human health/disease? Compare the two ideas. State one practical application.
29. How is ■ Human health/disease related to microbes and welfare.? Compare the two ideas. Justify your response.
30. How is microbes and welfare. related to ■ Human health/disease? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Human health/disease to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply microbes and welfare. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Human health/disease to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply microbes and welfare. to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply ■ Human health/disease to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply microbes and welfare. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Human health/disease to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply microbes and welfare. to a realistic situation. What would you do or calculate? State one practical application.
39. Apply ■ Human health/disease to a realistic situation. What would you do or calculate? Justify your response.
40. Apply microbes and welfare. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Human health/disease. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of microbes and welfare.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Human health/disease. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of microbes and welfare.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Human health/disease. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of microbes and welfare.. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of ■ Human health/disease. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of microbes and welfare.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Human health/disease. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of microbes and welfare.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Human health/disease. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show microbes and welfare.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Human health/disease. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show microbes and welfare.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Human health/disease. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show microbes and welfare.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Human health/disease. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show microbes and welfare.. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Human health/disease. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show microbes and welfare.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — BIOLOGY — Biotechnology

Coverage: ■ Principles/processes; recombinant DNA; applications; GMOs.

Understanding

1. What is ■ Principles/processes, and what are its main features? Use a simple example.
2. What is recombinant DNA, and what are its main features? Use a school-life example.
3. What is applications, and what are its main features? Use a real-world example.
4. What is GMOs., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Principles/processes, and what are its main features? Explain in your own words.
6. What is recombinant DNA, and what are its main features? Give a step-by-step response.
7. What is applications, and what are its main features? Mention one common misconception.
8. What is GMOs., and what are its main features? State one practical application.
9. What is ■ Principles/processes, and what are its main features? Justify your response.
10. What is recombinant DNA, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to applications and explain each. Use a simple example.
12. Give two examples related to GMOs. and explain each. Use a school-life example.
13. Give two examples related to ■ Principles/processes and explain each. Use a real-world example.
14. Give two examples related to recombinant DNA and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to applications and explain each. Explain in your own words.
16. Give two examples related to GMOs. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Principles/processes and explain each. Mention one common misconception.
18. Give two examples related to recombinant DNA and explain each. State one practical application.
19. Give two examples related to applications and explain each. Justify your response.
20. Give two examples related to GMOs. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Principles/processes related to recombinant DNA? Compare the two ideas. Use a simple example.
22. How is recombinant DNA related to applications? Compare the two ideas. Use a school-life example.
23. How is applications related to GMOs.? Compare the two ideas. Use a real-world example.
24. How is GMOs. related to ■ Principles/processes? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Principles/processes related to recombinant DNA? Compare the two ideas. Explain in your own words.
26. How is recombinant DNA related to applications? Compare the two ideas. Give a step-by-step response.
27. How is applications related to GMOs.? Compare the two ideas. Mention one common misconception.
28. How is GMOs. related to ■ Principles/processes? Compare the two ideas. State one practical application.
29. How is ■ Principles/processes related to recombinant DNA? Compare the two ideas. Justify your response.
30. How is recombinant DNA related to applications? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply applications to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply GMOs. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Principles/processes to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply recombinant DNA to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply applications to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply GMOs. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Principles/processes to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply recombinant DNA to a realistic situation. What would you do or calculate? State one practical application.
39. Apply applications to a realistic situation. What would you do or calculate? Justify your response.
40. Apply GMOs. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Principles/processes. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of recombinant DNA. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of applications. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of GMOs.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Principles/processes. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of recombinant DNA. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of applications. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of GMOs.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Principles/processes. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of recombinant DNA. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show applications. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show GMOs.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Principles/processes. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show recombinant DNA. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show applications. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show GMOs.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Principles/processes. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show recombinant DNA. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show applications. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show GMOs.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — BIOLOGY — Ecology

Coverage: ■ Organisms/populations; ecosystem; biodiversity; conservation.

Understanding

1. What is ■ Organisms/populations, and what are its main features? Use a simple example.
2. What is ecosystem, and what are its main features? Use a school-life example.
3. What is biodiversity, and what are its main features? Use a real-world example.
4. What is conservation., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Organisms/populations, and what are its main features? Explain in your own words.
6. What is ecosystem, and what are its main features? Give a step-by-step response.
7. What is biodiversity, and what are its main features? Mention one common misconception.
8. What is conservation., and what are its main features? State one practical application.
9. What is ■ Organisms/populations, and what are its main features? Justify your response.
10. What is ecosystem, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to biodiversity and explain each. Use a simple example.
12. Give two examples related to conservation. and explain each. Use a school-life example.
13. Give two examples related to ■ Organisms/populations and explain each. Use a real-world example.
14. Give two examples related to ecosystem and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to biodiversity and explain each. Explain in your own words.
16. Give two examples related to conservation. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Organisms/populations and explain each. Mention one common misconception.
18. Give two examples related to ecosystem and explain each. State one practical application.
19. Give two examples related to biodiversity and explain each. Justify your response.
20. Give two examples related to conservation. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Organisms/populations related to ecosystem? Compare the two ideas. Use a simple example.
22. How is ecosystem related to biodiversity? Compare the two ideas. Use a school-life example.
23. How is biodiversity related to conservation.? Compare the two ideas. Use a real-world example.
24. How is conservation. related to ■ Organisms/populations? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Organisms/populations related to ecosystem? Compare the two ideas. Explain in your own words.
26. How is ecosystem related to biodiversity? Compare the two ideas. Give a step-by-step response.
27. How is biodiversity related to conservation.? Compare the two ideas. Mention one common misconception.
28. How is conservation. related to ■ Organisms/populations? Compare the two ideas. State one practical application.
29. How is ■ Organisms/populations related to ecosystem? Compare the two ideas. Justify your response.
30. How is ecosystem related to biodiversity? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply biodiversity to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply conservation. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Organisms/populations to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ecosystem to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply biodiversity to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply conservation. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Organisms/populations to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply ecosystem to a realistic situation. What would you do or calculate? State one practical application.
39. Apply biodiversity to a realistic situation. What would you do or calculate? Justify your response.
40. Apply conservation. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Organisms/populations. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of ecosystem. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of biodiversity. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of conservation.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Organisms/populations. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ecosystem. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of biodiversity. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of conservation.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Organisms/populations. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of ecosystem. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show biodiversity. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show conservation.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Organisms/populations. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show ecosystem. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show biodiversity. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show conservation.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Organisms/populations. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ecosystem. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show biodiversity. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show conservation.. Include the relevant rule or principle.